

OTP Attack Analysis Report

Brute Force vs Collision-Based Attacks on Time-Limited OTPs

Generated: 2026-05-25T20:07:44

Table of Contents

1. Mathematical Foundations
2. Threat Model
3. Attack Model
4. Security Implications
5. Brute Force Scatter Plot
6. Collision Scatter Plot
7. Brute Force Surface Plot
8. Collision Surface Plot
9. Log-Scale Plots
10. Side-by-Side Comparisons
11. GPU/Chunking Notes
12. Conclusions & Future Exploration

Mathematical Foundations

Keyspace: $62^{10} = 8.393e+17$

Brute-force probability $p_{bf} = 1.191e-18$

Weak entropy bits = 40

Weak space size = $1.100e+12$

Overlap factor = 0.1

Collision probability $p_{coll} = 9.095e-14$

Cumulative probability: $P(k) = 1 - (1 - p)^k$.

Threat Model

Attacker: unlimited attempts, automation, knowledge of OTP timing and format.

Defender: relies solely on OTP, no rate limiting, no MFA, no anomaly detection.

Attack Model

Brute force: uniform sampling over 62^{10} .

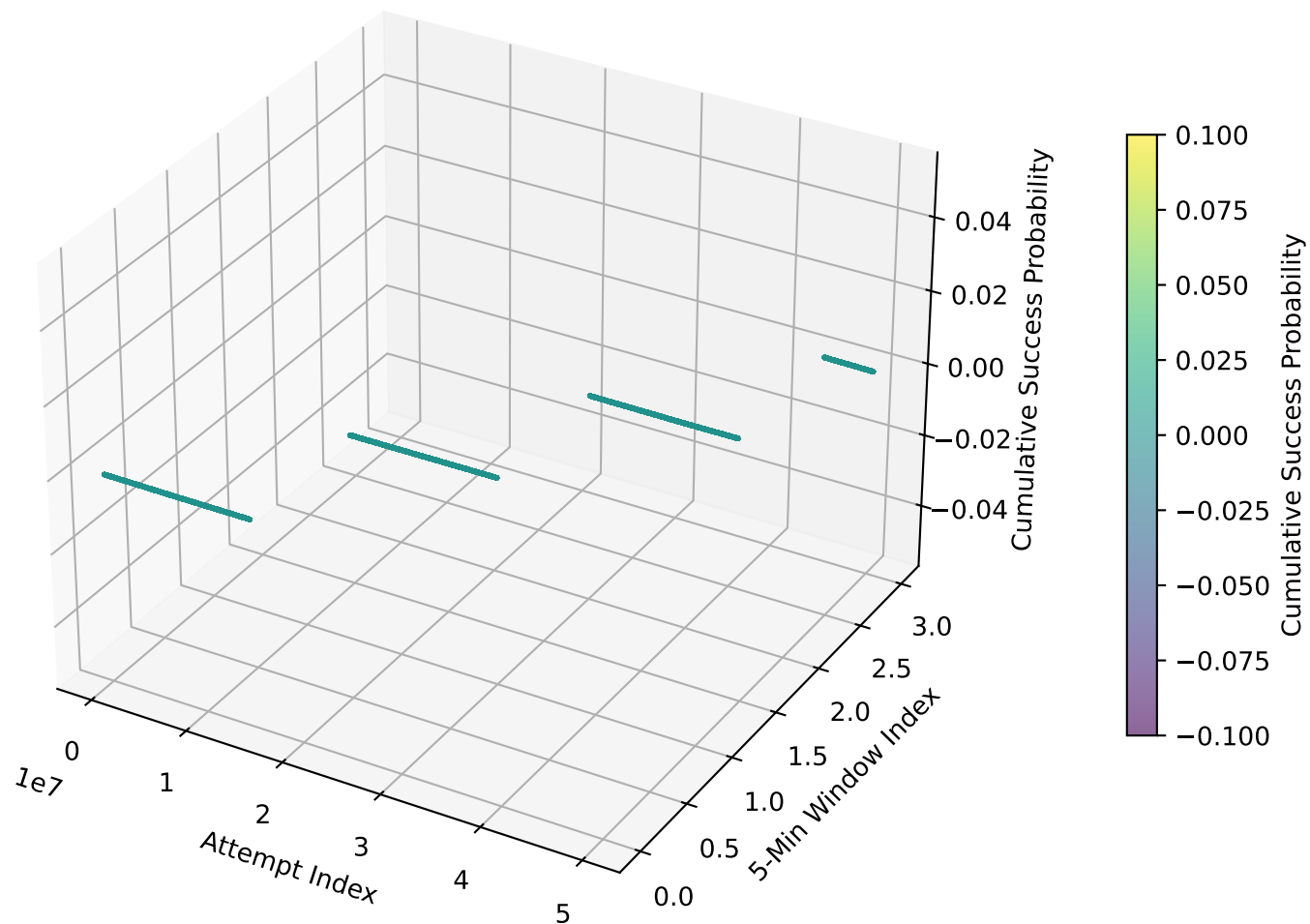
Collision: attacker exploits weak entropy (E bits) and overlap with true OTP distribution.

Security Implications

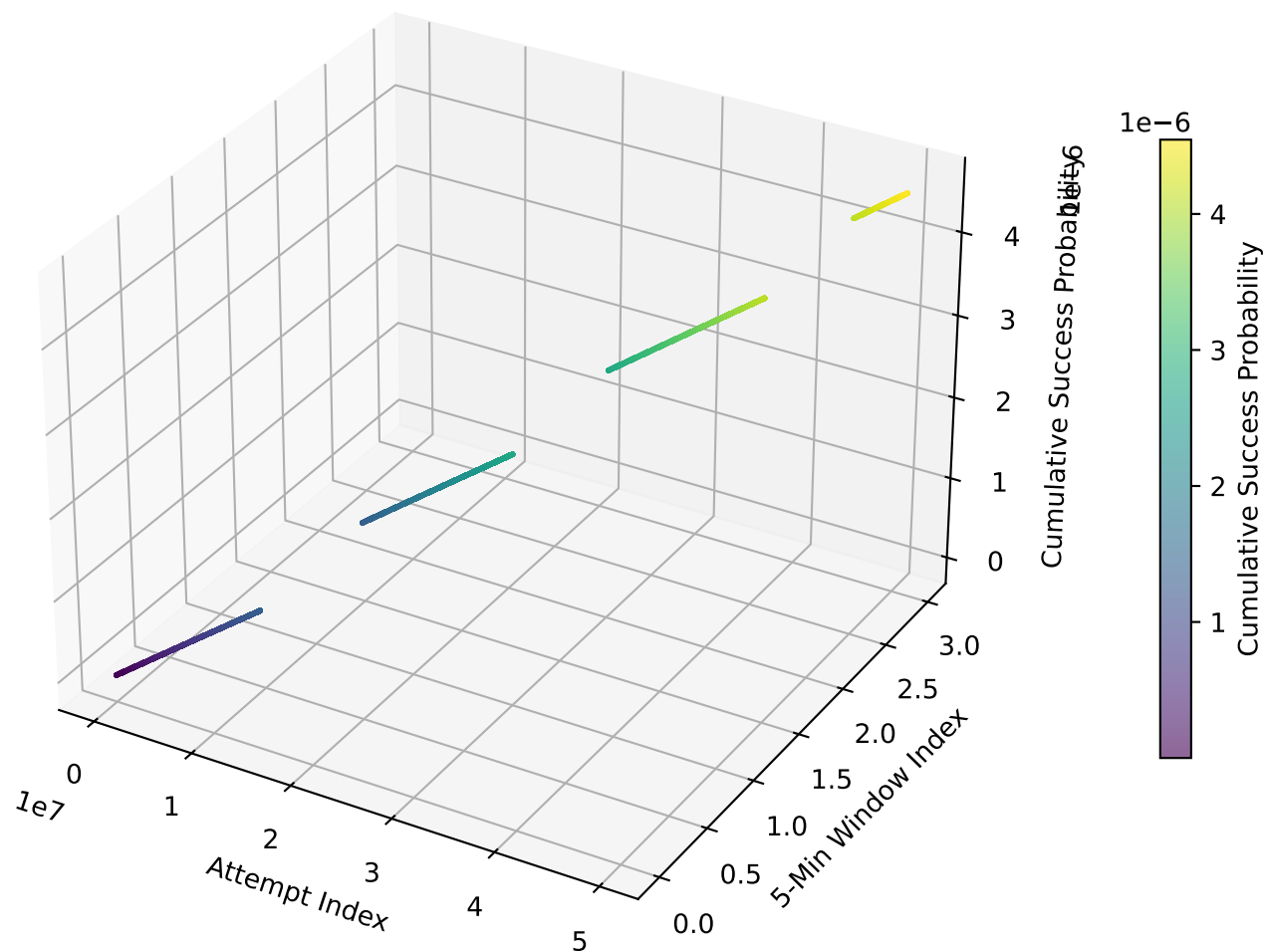
Weak or biased OTP generators drastically increase attack success.

Single-factor OTP authentication is unsafe without rate limiting and MFA.

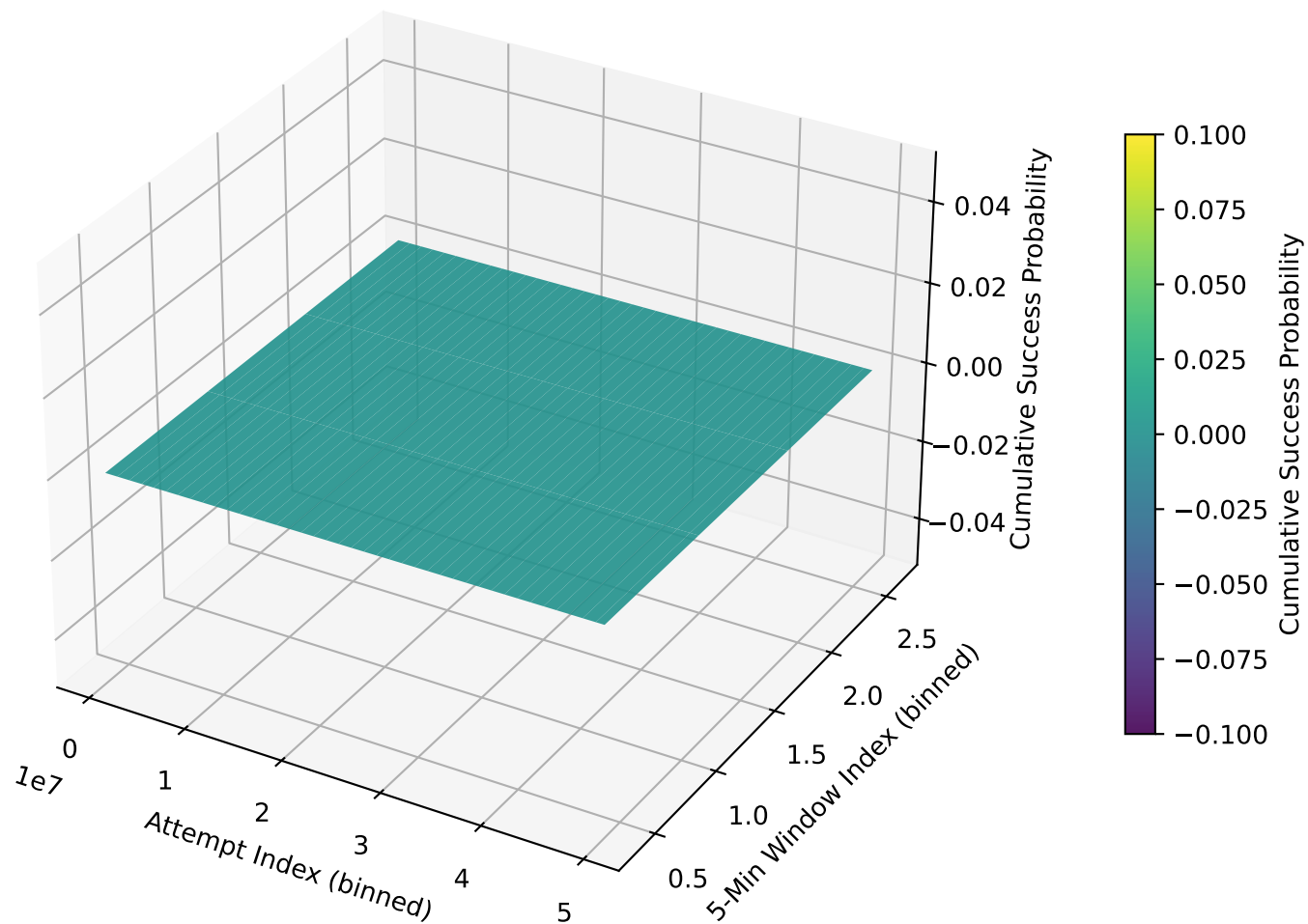
Brute Force Attack Scatter (Downsampled)



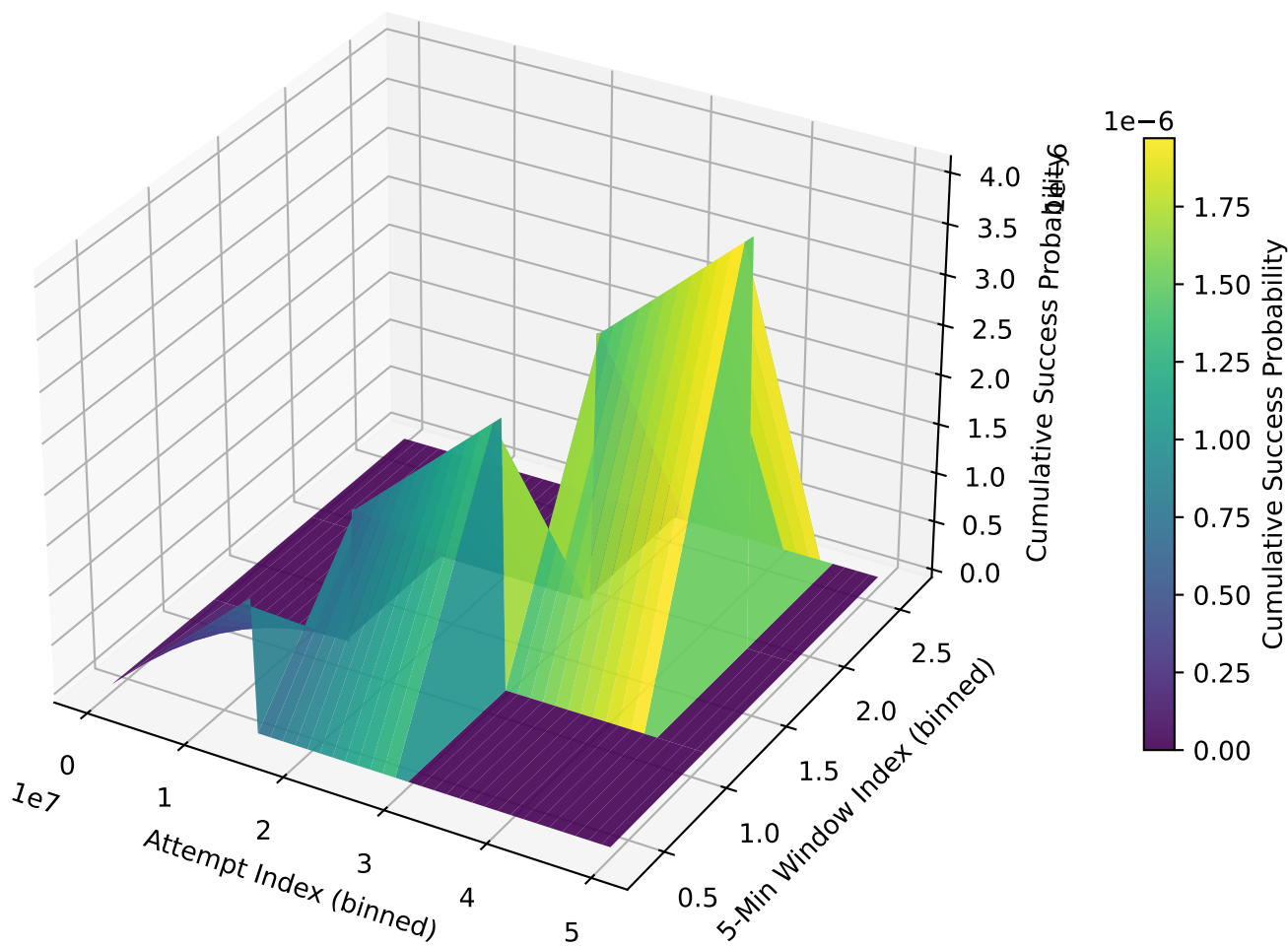
Collision Attack Scatter (Downsampled)



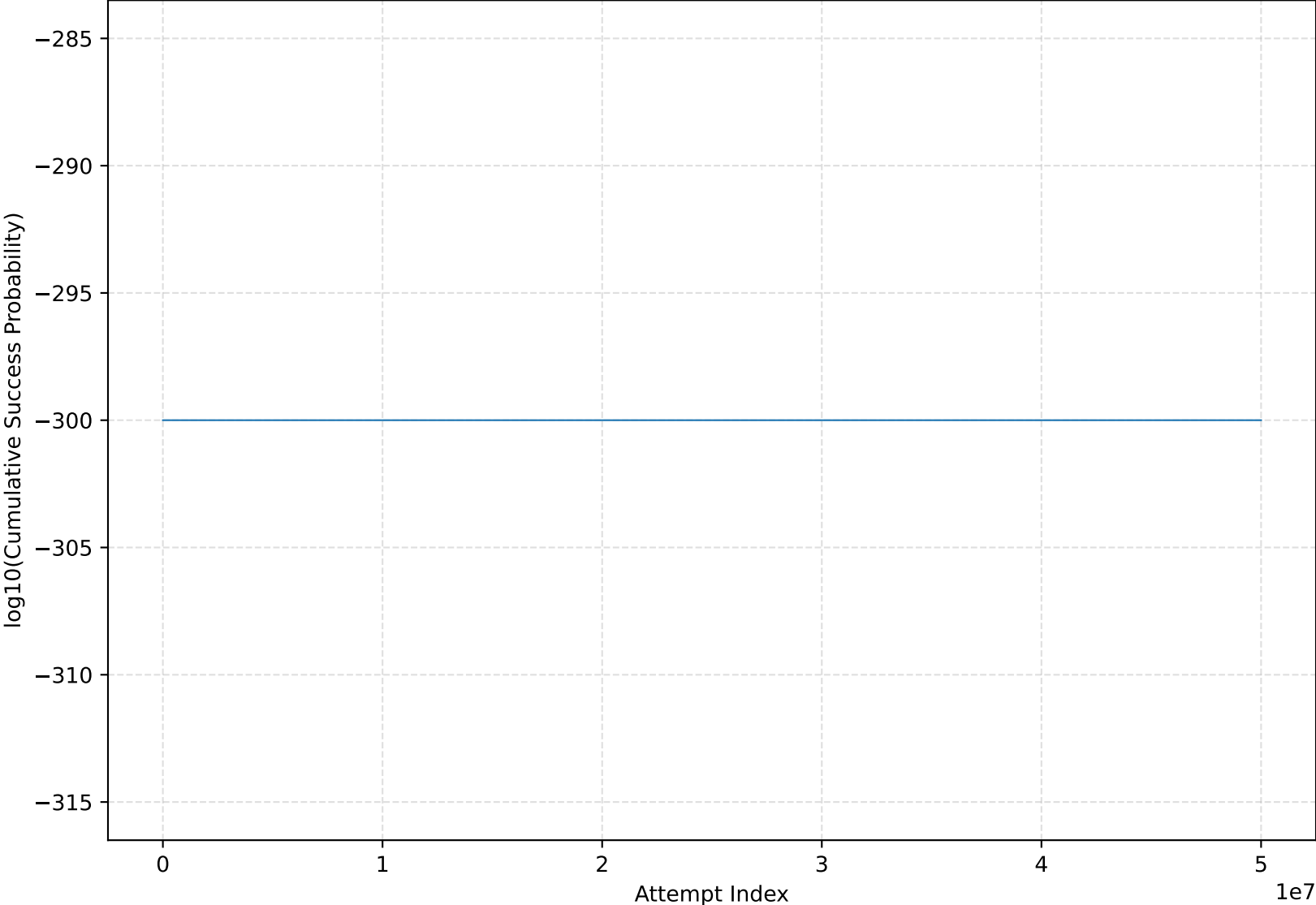
Brute Force Success Probability Surface



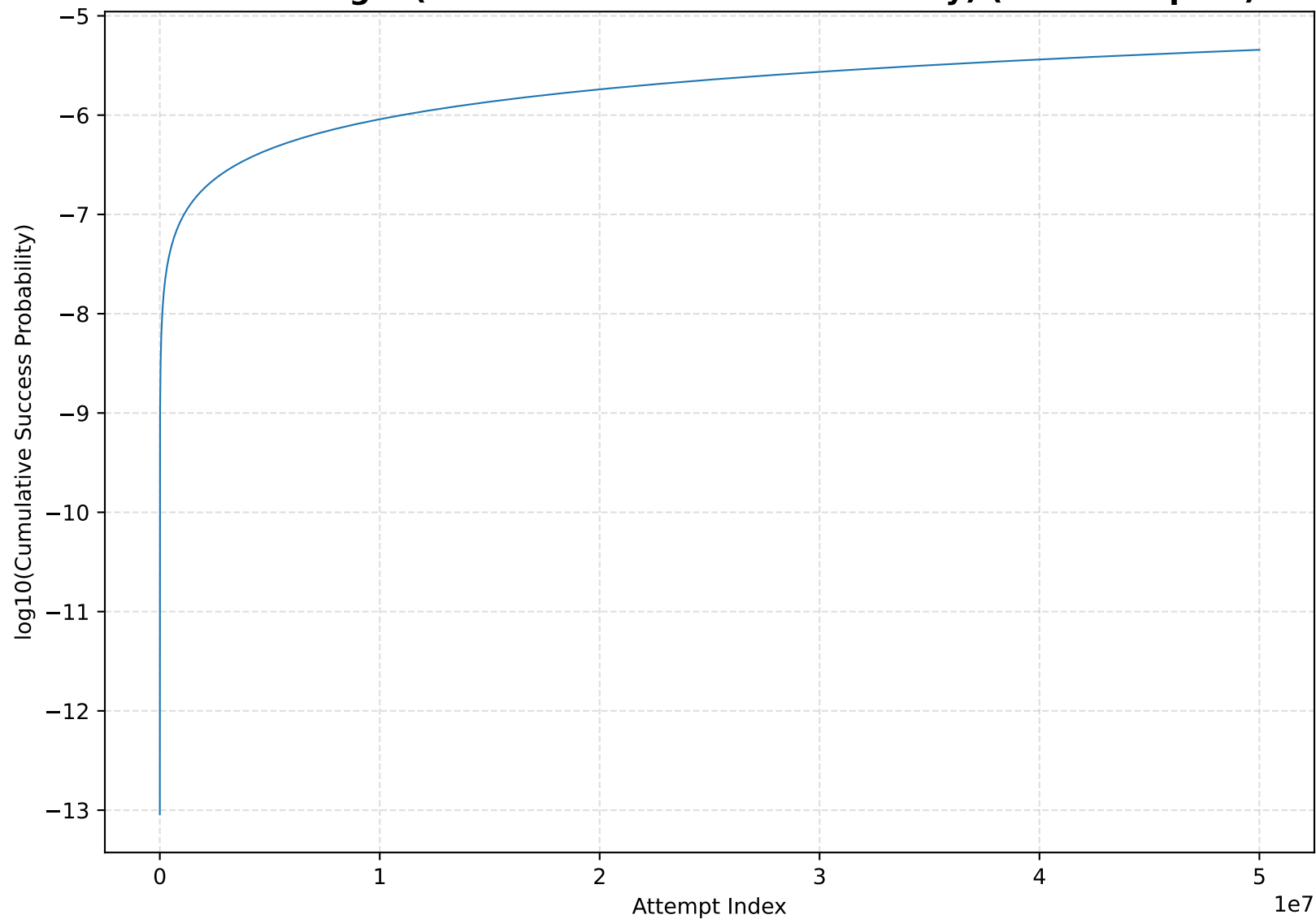
Collision Success Probability Surface



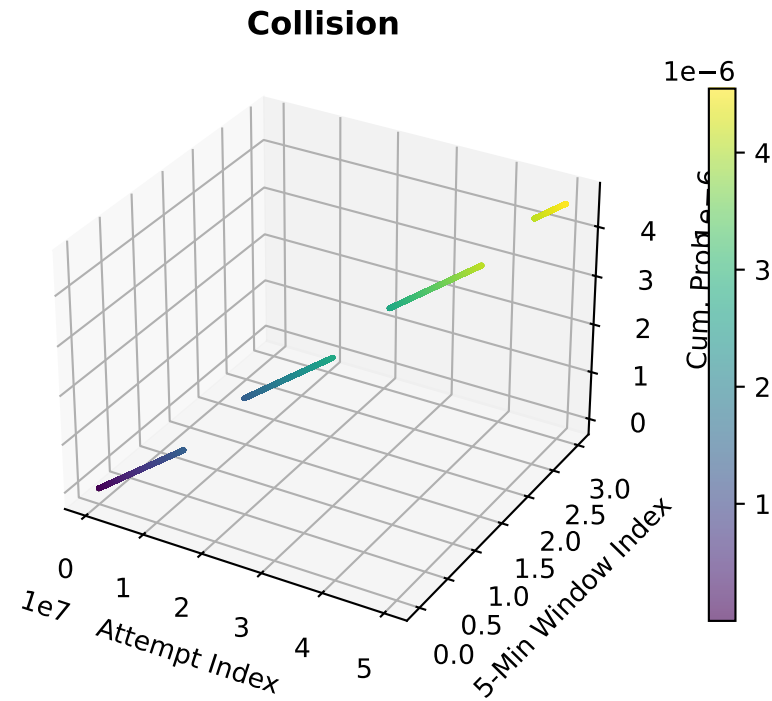
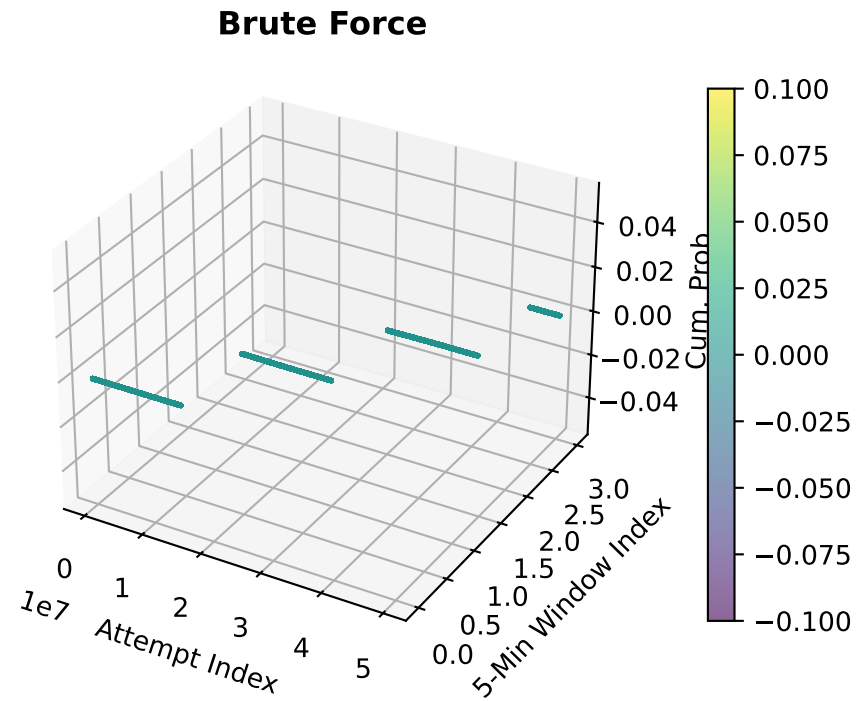
Brute Force log10(Cumulative Success Probability) (Downsampled)



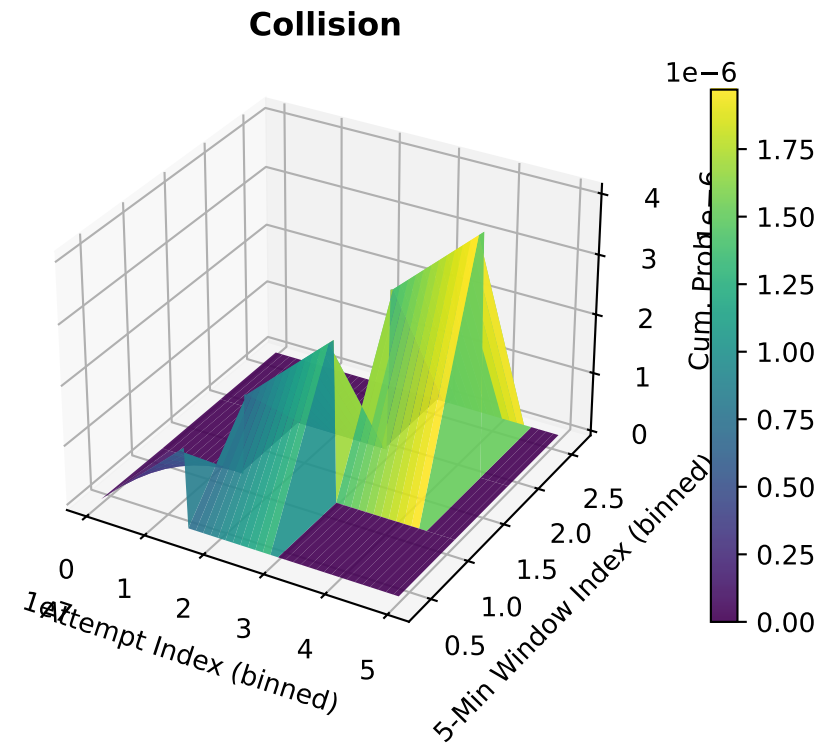
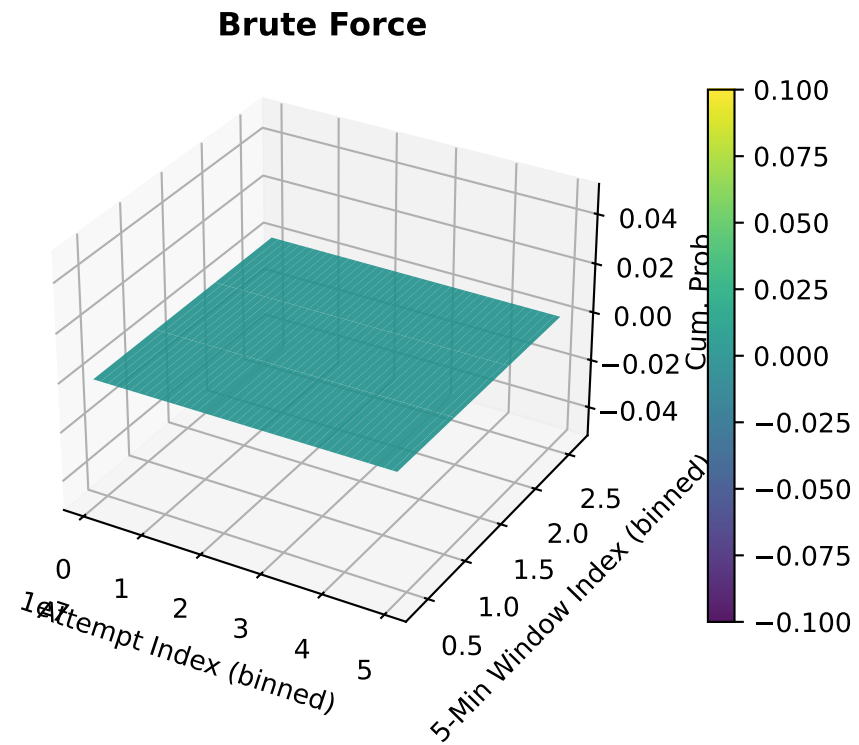
Collision $\log_{10}(\text{Cumulative Success Probability})$ (Downsampled)



Brute Force vs Collision (Scatter, Downsampled)



Brute Force vs Collision (Surface)



GPU and Chunking Notes

GPU acceleration enabled: False.

Cumulative probabilities are computed in chunks to avoid excessive memory usage.

If CuPy is installed and a compatible GPU is available, chunks are computed on GPU.

Otherwise, the script transparently falls back to NumPy on CPU.

Logical total attempts modeled: 50,000,000.

Scatter/log plots are downsampled to at most 200,000 / 200,000 points.

Conclusions & Future Exploration

Single-factor OTP authentication is insufficient for high-value accounts.

Weak entropy or biased generators can make collision-style attacks far more effective than naive brute force.

Mitigations: strong RNGs, rate limiting, MFA, device binding, anomaly detection.